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RESEARCH

A resting-state functional magnetic resonance imaging meta-analysis of differences in brain activity between children and adolescents with attention-deficit/hyperactivity disorder using activation likelihood estimation

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Abstract

Attention-deficit/hyperactivity disorder (ADHD) is a prevalent neurodevelopmental disorder that often persists from childhood into adolescence and adulthood. Resting-state functional magnetic resonance imaging (rs-fMRI) provides valuable insights into the intrinsic neural activity associated with ADHD. However, despite increasing neuroimaging research, the developmental specificity of spontaneous alterations in brain activity in children and adolescents with ADHD remains poorly understood. A comprehensive activation likelihood estimation (ALE) meta-analysis was performed on rs-fMRI data to investigate alterations in spontaneous brain activity in children and adolescents with ADHD compared with healthy controls (HCs). A contrast analysis was conducted to assess potential overlap in altered brain regions between the child and adolescent ADHD groups. The robustness of the findings was evaluated using a jackknife sensitivity analysis. A systematic review of the literature identified 28 rs-fMRI studies (1019 ADHD patients and 943 HCs). In children with ADHD, ALE revealed decreased spontaneous neural activity in the left middle frontal gyrus, superior frontal gyrus, medial frontal gyrus, precentral gyrus, and subgyral region, with no regions showing increased activity. In adolescents with ADHD, increased activity was observed in the bilateral paracentral lobule, left postcentral gyrus, and medial frontal gyrus, whereas decreased activity was found in the cerebellar tonsil, uvula, declive, anterior lobe, and superior/medial frontal gyrus. No significant clusters were identified in the contrast analyses. The jackknife sensitivity analysis demonstrated robustness in 9 of 17 iterations for children and 4 of 5 iterations for adolescent-specific cerebellar findings. Spontaneous alterations in brain activity in children and adolescents with ADHD reflect developmentally distinct neural mechanisms and may guide future age-specific neuroimaging research.

Keywords Attention-deficit/hyperactivity disorder · Children · Adolescent · Resting-state functional magnetic resonance imaging · Spontaneous brain activity

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Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a highly prevalent and heterogeneous neurodevelopmental disorder that is driven primarily by polygenic genetic influences and characterized by structural and functional abnormalities within corticostriatal circuits. These neural alterations are closely linked to deficits in attention, impulse control, and executive function [1]. ADHD commonly manifests during early childhood and often continues into adolescence and adulthood, with a global prevalence estimated at approximately 6–7% in pediatric populations and 4–5% among adults [2]. Despite recent advances in the delineation of the roles of large-scale brain networks and neuroanatomical mechanisms in the pathophysiology of ADHD [3, 4], further investigation into age-specific neurofunctional alterations remains essential to elucidate its developmental trajectory and heterogeneity.

Researchers have hypothesized that neurodevelopmental abnormalities in patients with ADHD reflect a delay in brain maturation, which contributes to impairments in late-developing cognitive functions [5, 6]. Neuroimaging evidence has repeatedly shown atypical development across large-scale brain systems in patients with ADHD, with consistent disruptions observed in the prefrontal cortex, basal ganglia, and cerebellum [7]. These abnormalities are reflected in compromised white matter integrity and altered patterns of functional connectivity, underscoring widespread structural and functional dysregulation in patients with the disorder.

Resting-state functional magnetic resonance imaging (rs-fMRI) is considered more suitable than task-based fMRI for pediatric populations because it allows the assessment of intrinsic brain activity under natural, task-free conditions [8]. A range of voxel-level indices, including amplitudes of low-frequency fluctuations (ALFF), dynamic ALFF (dALFF), fractional ALFF (fALFF), and regional homogeneity (ReHo), are routinely applied in neuroimaging studies to capture intrinsic fluctuations in resting-state brain activity [9]. ALFF and fALFF capture the strength of slow oscillatory activity linked to local neuronal excitability, whereas dALFF extends this framework by quantifying temporal variability in ALFF signals, thereby reflecting dynamic fluctuations in regional activity over time. Moreover, ReHo assesses the local synchronization of blood oxygen level-dependent (BOLD) signal time series among neighboring voxels, offering a measure of regional functional coherence [10]. Rs-fMRI has become an essential modality for investigating intrinsic brain activity in individuals with ADHD. Previous studies using ALFF, fALFF, dALFF, and ReHo have reported aberrant spontaneous activity across multiple brain regions, including the prefrontal cortex, sensorimotor areas, parietal lobe, basal ganglia, and cerebellum [11–38]. These alterations are believed to reflect the neurophysiological basis of core ADHD symptoms such as inattention, impulsivity, and hyperactivity. However, although many studies have reported resting-state functional abnormalities in both children and adolescents with ADHD, the findings have been inconsistent and remain a subject of debate.

Given the heterogeneity of previous findings, a quantitative meta-analytic framework is warranted to identify consistent and replicable patterns of altered spontaneous brain activity in individuals with ADHD. Activation likelihood estimation (ALE) offers a robust coordinate-based approach to synthesize voxel-level neuroimaging results across independent studies, thereby increasing statistical power and mitigating interstudy variability [39]. ALE provides a rigorous quantitative framework that synthesizes findings across independent studies, allowing the identification of spatially convergent brain regions and yielding more reliable inferences about functional alterations under specific conditions. Nevertheless, Samea et al. reported no significant convergence across structural and functional alterations in individuals with ADHD, possibly because of the inclusion of heterogeneous imaging modalities and the lack of restriction to rs-fMRI data alone [40]. Lukito et al. utilized seed-based mapping (SDM) to examine structural alterations and observed reduced gray matter volume in the ventromedial orbitofrontal cortex of ADHD patients; however, their study did not investigate alterations in spontaneous brain activity during the resting state [41]. Chen et al. identified overlapping and modality-specific abnormalities in the orbitofrontal cortex and other regions without age stratification and using SDM rather than ALE [42]. Sutcubasi et al. conducted a seed-based resting-state meta-analysis and reported default mode network (DMN) dysconnectivity in ADHD patients but did not assess spontaneous activity using ALE [43], and Cortese et al. did not observe a consistent connectivity convergence in an ALE meta-analysis [44].

We addressed these gaps by conducting an ALE meta-analysis to examine alterations in spontaneous brain activity in children and adolescents with ADHD. We hypothesized that, compared with healthy controls (HCs), children and adolescents with ADHD would exhibit distinct, age-specific patterns of spontaneous neural activity. Additionally, contrast analyses were conducted to examine shared versus distinct regional abnormalities across age groups. The findings of this meta-analysis may provide novel insights into the age-dependent neurofunctional mechanisms of ADHD and a foundation for more developmentally informed diagnostic and intervention strategies.

Materials and methods

We preregistered the study protocol on PROSPERO on April 12, 2024 (CRD42024531049) to promote transparency and minimize the risk of bias. A summary of any deviations in the protocol is provided in the Supplementary Table 7.

Literature search

Following the PRISMA guidelines [45], we systematically searched four major electronic databases, namely, PubMed, Web of Science, Embase, and CNKI, for studies published up to December 19, 2024. The search strategy included the following keywords: ("attention deficit hyperactivity disorder" OR "ADHD") AND ("adolescent" OR "child" OR "children" OR "teenager" OR "youth" OR "pediatric") AND ("resting-state functional magnetic resonance imaging" OR "rs-fMRI" OR "functional MRI" OR "fMRI") AND ("amplitude of low-frequency fluctuations" OR "ALFF" OR "low frequency fluctuation" OR "regional homogeneity" OR "ReHo" OR "homogeneity" OR "brain activity"). We screened the reference lists of relevant studies to identify potentially missed studies and minimize omissions.

Inclusion and exclusion criteria

Studies were considered eligible for inclusion if they met the following criteria: (1) ADHD was diagnosed according to the DSM-IV or DSM-5; (2) rs-fMRI studies of ADHD patients using ReHo, ALFF, fALFF, or dALFF; (3) a case–control design was employed to assess differences in resting-state neural activity between ADHD patients and HCs; and (4) whole-brain analysis results were reported in standardized stereotaxic space (Montreal Neurological Institute (MNI) or Talairach).

Studies were excluded if they (1) utilized functional connectivity, independent component analysis, or degree centrality; (2) utilized task-based fMRI (t-fMRI) studies on ADHD patients or cerebral perfusion assessments; (3) were literature reviews, systematic reviews, meta-analyses, case reports, letters, conference abstracts, or editorials; (4) did not report complete stereotactic coordinates (x, y, z); or (5) lacked an HC group. In cases of overlapping datasets, the publication with the largest sample size and most rigorous methodological detail was prioritized to preserve data independence.

Data extraction

Two authors (SYP and HYZ) separately performed data extraction based on a standardized protocol and predefined data collection form. Discrepancies were resolved through discussion and consensus. The extracted variables included key study characteristics, such as the first author, publication year, number of participants, sample size, mean age, sex distribution, intelligence quotient (IQ), ADHD subtype, diagnostic criteria, and medication status. Details of the rs-fMRI data included the scanning equipment, applied methods, the number of differentially altered brain regions between ADHD patients and healthy controls, corrective methods, and standard coordinate systems used for reporting the results. All coordinates were extracted independently based on the methodological requirements of an ALE meta-analysis.

Consistent with previous studies [46], we applied an age threshold of 12 years to distinguish adolescents from children. We categorized the populations in the studies into child (< 12 years) and adolescent (≥ 12 years) groups based primarily on the reported mean ± standard deviation (SD) of participant age to explore developmental differences. If the mean ± SD clearly placed the sample in one group, the study was classified accordingly, regardless of whether the age range spanned both groups. When the mean ± SD was not available, classification was based on the minimum age reported. If neither criterion allowed clear categorization and if the age range spanned both groups (e.g., Yang et al., 2011), the study was excluded from the age-specific subgroup analyses. This approach enabled a more developmentally informed analysis of age-specific neural alterations in ADHD patients.

Quality and risk of bias assessments

To evaluate methodological rigor, The Newcastle–Ottawa Quality Assessment Scale (NOS), which is a standardized instrument for evaluating nonrandomized studies, was used to evaluate methodological rigor [47]. Based on total scores, the risk of bias was categorized as low (7–9), moderate (4–6), or high (< 4) [48].

Additionally, we referenced the domains outlined in the Risk of Bias 2.0 (ROB 2.0) tool developed by the Cochrane Collaboration, including participant selection, deviations from the intended interventions, missing data, outcome measurements, and selection of reported results. Each domain was rated as "low risk", "some concerns", or "high risk", and an overall risk judgment was assigned accordingly. Quality assessments were independently performed by two authors (SYP and HYZ), with any discrepancies resolved through discussion with additional team members.

ALE analysis

This meta-analysis applied the ALE approach to examine age-specific alterations in spontaneous brain activity between children and adolescents with ADHD.

Single-dataset analysis

In this study, the ALE algorithm was implemented using the GingerALE 2.3 platform [49]. Initially, the Lancaster (SPM) algorithm available in GingerALE was used to convert all Talairach coordinates into MNI space [50]. Gaussian function smoothing with a full width at half maximum (FWHM) was then applied based on the sample size of each study group to model the reported coordinates of activated brain regions into three-dimensional (3D) brain masks, creating modeled activation (MA) maps. Next, a single-dataset ALE analysis was performed separately for the child and adolescent ADHD groups. ALE scores were calculated by combining the MA maps across studies within each group to identify regions showing consistent activation patterns. These group-specific ALE maps were then subjected to statistical testing. Statistical significance (P values) was calculated to generate a 3D-P value image. This study employed cluster-level familywise error (cFWE) correction, which was set at $P < 0.05$, and the threshold permutations were set at $P < 0.05$ with 1,000 permutations to control for multiple comparisons. The threshold was determined using a three-dimensional distribution of P values, and the resulting ALE map reflected regions that met this corrected significance level [51, 52].

Contrast dataset analysis

According to the protocol, we intended to perform contrast analyses using the ALE method to assess potential developmental differences between children and adolescents with ADHD. We also initially considered conducting contrast analyses limited to male-only samples, given the higher prevalence of ADHD among males and the known sex-related clinical differences. However, due to an insufficient number of studies of children exclusively involving males, the sex-specific contrast analysis was not performed. In accordance with the ALE manual, a contrast analysis was performed. First, single-dataset ALE analyses were conducted separately for the child and adolescent groups. We subsequently performed two contrast analyses to assess direction-specific age-related differences: (1) child ADHD < adolescent ADHD and (2) child ADHD >adolescent ADHD. Based on these results, the joint activation probability between the two groups was calculated to identify brain regions that showed convergent activation across both age groups. Simultaneously, significant differences in convergence between the datasets were assessed. Statistical significance was determined using 1,000 permutations, with the threshold set at a P value less than 0.05 [53]. Finally, a conjunction image was generated to display the overlapping activation between the two datasets. The final coordinates were mapped onto a standard brain template (Colin) using Mango software [54].

Sensitivity analysis

A jackknife sensitivity analysis was conducted using Ginger ALE 2.3 software to assess the replicability of the results. Each study was sequentially excluded, and an ALE meta-analysis was performed on the remaining data to examine the impact on the consistency of the results.

Results

Included studies and sample characteristics

A total of 2,730 records were initially identified, and one additional study was included through the manual reference screen. Ultimately, 28 rs-fMRI studies met the inclusion criteria (Fig. 1) [11–38], were published between 2007 and 2025, and involved participants aged approximately 8 to 13 years. Among the 28 studies, 12 included only males and 16 included mixed-sex samples. The methods used were ReHo (14 studies), ALFF (14 studies), fALFF (3 studies), and dALFF (2 studies), with 5 studies using ALFF and ReHo. In total, 1962 participants (1019 ADHD patients and 943 HCs) contributed 217 foci for the ALE meta-analysis (Table 1, Table S1).

Twenty-two studies were categorized as examining childhood ADHD, and 5 studies were categorized as assessing adolescent ADHD (Tables S2–S3). One study by Yang [15] (participants aged 7–13 years) included both groups and was included only in the overall analysis to avoid classification bias.

ALE meta-analysis

Single-dataset analysis

We first conducted separate single-dataset meta-analyses of studies involving children and adolescents (Fig. 2). In the child group (Table 2), the ALE analysis did not identify any brain regions showing greater activation in children with ADHD than in HCs. However, decreased activity was detected mainly in the left frontal lobe, including the middle frontal gyrus (Brodmann areas [BAs] 6 and 9), superior

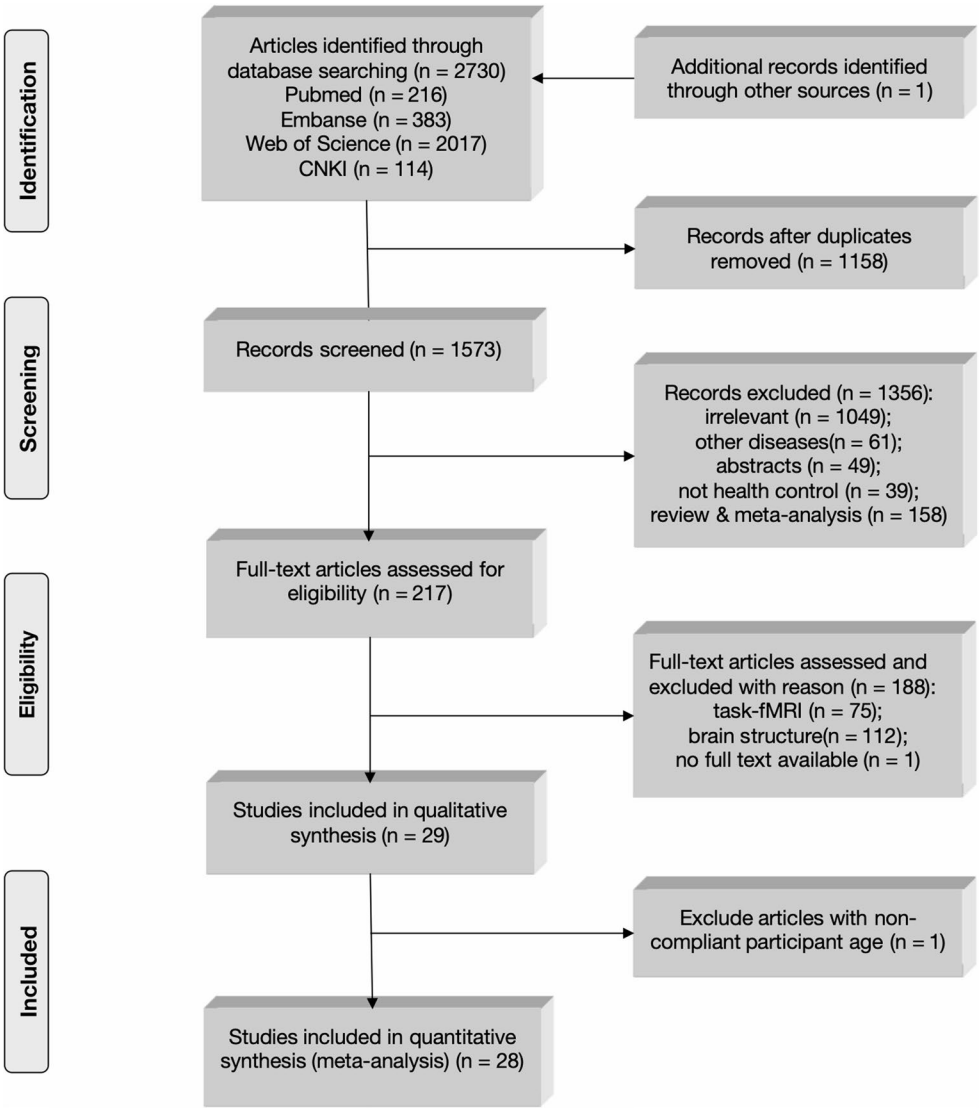


Fig. 1 PRISMA flow diagram of the study selection process

frontal gyrus (BA 6), medial frontal gyrus (BA 6), precentral gyrus (BA 9), and subgyral region (BA 6). Second, in the adolescent group (Table 3), increased activation was observed in the paracentral lobule (BA 4 and 5), postcentral gyrus (BA 3), and medial frontal gyrus (BA 6), mainly in sensorimotor-related regions. In contrast, decreased activation was observed primarily in the cerebellum, including the tonsil, uvula, declive, anterior lobe, and nodule, as well as in the superior and medial frontal gyri (BA 9 and 10). No spatial overlap was observed among the significant clusters identified in the child and adolescent groups (Fig. 3).

Finally, when children and adolescents with ADHD were analyzed as a single group and compared with healthy controls, the ALE analysis revealed no significant clusters of altered brain activity.

Contrast analysis

For the contrast child<adolescent ADHD, a total of 24 experiments contributed 117 foci. No significant clusters were identified. For the contrast child>adolescent ADHD, 26 experiments contributed 125 foci. No significant clusters were identified in this comparison (Table 4).

Sensitivity analysis

The jackknife sensitivity analysis of decreased brain activity in children with ADHD compared with HCs revealed that all identified clusters were moderately replicated in at least 9 of 17 iterations (Table S4). In adolescents with ADHD, decreased activity in the cerebellar tonsil, declive of the cerebellar posterior lobe, the anterior lobe of the cerebellum, the nodule and the uvula of the anterior lobe of the cerebellum, and the superior frontal gyrus was consistently

Table 1 Characteristics of the included studies

Author, year	Sample size		Age group	IQ Score		Group Difference (Age, IQ)	Sex	MRI equipment and Field strength	Medication Status	Diagnostic Criteria	Subtype	Method	Foci (*n*)	Correction Methods	Coordinate System
	Patients	HCs		Patient	HCs										
Cao QJ, 2007 [11]	15	15	Child	>80	>80	N, Y	M	Siemens Trio 3.0 T	N/A	DSM-IV	ADHD-I/ADHD-C	ReHo	14	AlphaSim $P < 0.05$	Talairach
Zang YF, 2007 [12]	13	12	Adolescent	99.0±11.6	118±11.7	N, Y	M	Siemens Trio 3.0 T	stimulant-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF	11	Puncor<0.01	Talairach
Pang GF, 2009 [13]	15	14	Child	>80	>80	N, N/A	M	Siemens Avto 1.5 T	stimulant-naïve	DSM-IV	N/A	ReHo	11	Puncor<0.01	Talairach
Yang H, 2011 [14]	17	17	Child	89.8±10.3	117.5±13.6	N, N	M	GE 3.0 T	medication-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF	5	AlphaSim $P<0.05$	Talairach
Yang ZL, 2011 [15]	30	33	Not grouped	>80	>80	N, Y	M	Siemens Trio 3.0 T	medication-naïve	DSM-IV	N/A	fALFF	11	Puncor<0.01	Talairach
Wu DQ, 2012 [16]	32	30	Child	>80	>80	N, N/A	M and F	Siemens Trio 3.0 T	stimulant-naïve	DSM-IV	ADHD-C	ALFF	10	AlphaSim $P<0.01$	Talairach
An L, 2013 [17]	19	23	Adolescent	102.7±10.4	113.5±12.8	N, Y	M	Siemens Trio 3.0 T	medication-naïve	DSM-IV	ADHD-I/ADHD-HI	ALFF and ReHo	10 and 19	AlphaSim $P < 0.01$	MNI
An L, 2013 [18]	22	30	Adolescent	110.5±12.5	121.0±12.7	N, Y	M	Siemens Trio 3.0 T	stimulant-naïve	DSM-IV	ADHD-I/ADHD-HI	ReHo	9	AlphaSim $P < 0.05$	MNI
Yang RX, 2013 [19]	18	18	Child	≥80	≥80	N, N/A	M and F	N/A	stimulant-naïve	DSM-IV	N/A	ALFF and ReHo	4 and 8	AlphaSim $P<0.05$	Talairach
Wang W, 2013 [20]	57	59	Child	106.0±15.5	113.7±14.6	N, N/A	M and F	Siemens Trio 3.0 T	stimulant-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF	7	AlphaSim $P<0.01$	Talairach
Li F,2014 [21]	33	32	Child	≥90	≥90	N, N	M	GE Signa 3.0 T	medication-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF	5	FWE $P<0.05$	Talairach
Yang RX, 2014 [22]	18	18	Child	95.77±10.79	100.93±8.33	N, N/A	M and F	GE Signa 1.5 T	drug-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF and ReHo	4 and8	AlphaSim $P<0.05$	Talairach
Wang SH, 2014 [23]	29	33	Child	102.48±15.76	107.10±9.12	N, N/A	M and F	Siemens Trio 1.5 T	medication-naïve	DSM-IV	ADHD-C	ALFF	3	FDR $P<0.05$	MNI
Yu X, 2015 [24]	30	30	Child	106.2±14.5	121.1±13.9	N, Y	M	Siemens Trio 3.0 T	drug-naïve	DSM-IV	ADHD-I/ADHD-HI	ReHo	6	AlphaSim $P<0.05$	MNI
Zhang XY, 2015 [25]	43	35	Child	116.13±17.00	119.17±15.78	N, N	M and F	GE Signa 3.0 T	drug-naïve	DSM-IV	ADHD-I/ADHD-C	ALFF	23	GRF $P<0.05$	MNI
Xie N, 2016 [26]	10	10	Child	≥80	≥80	N, N/A	M	GE Signa 1.5 T	N/A	DSM-IV	N/A	ALFF	5	GRF $P<0.05$	MNI
Wang JB, 2017 [27]	91	73	Adolescent	106	116	N, Y	M	GE Signa 3.0 T	N/A	DSM-IV	N/A	ALFF and ReHo	2 and 5	AlphaSim $P<0.05$	MNI
Zhao D, 2017 [28]	26	40	Child	>80	>80	N/A, N/A	M and F	Siemens Trio 3.0 T	N/A	DSM-IV	ADHD-C	ReHo	10	GRF $P<0.05$	MNI
Wang YF, 2017 [29]	38	42	Child	≥80	≥80	N, Y	M and F	Siemens Trio 3.0 T	N/A	DSM-IV	All subtypes	fALFF	6	AlphaSim $P<0.05$	MNI
Kim JI, 2018 [30]	67	44	Child	107.76±15.06	113.07±12.16	Y, N	M and F	Siemens Trio 3.0 T	drug-naïve	DSM-IV	All subtypes	ReHo	1	FWE $P=0.03$	MNI
Zhou JX, 2018 [31]	45	26	Child	114.00±13.27	121.00±13.87	N, N	M and F	GE Signa HDX 3.0 T	medication-naïve	DSM-5	N/A	ReHo	1	FWE $P<0.05$	MNI
Lou F, 2021 [32]	50	28	Child	116.94±17.45	122.43±13.84	N, N	M and F	GE Signa HDX 3.0 T	drug-naïve	DSM-5	N/A	dALFF	2	GRF $P<0.05$	MNI
Shang CY, 2021 [33]	96	114	Child	104.7±11.6	109.6±12.3	Y, Y	M and F	Siemens Magnetom Trio 3.0 T	drug-naïve	DSM-IV	N/A	ReHo	1	FDR $P<0.05$	MNI
Feng Y, 2023 [34]	52	51	Child	105.38±13.57	123.81±16.32	N, Y	M and F	GE Discovery MR750 3.0 T	drug-naïve	DSM-IV	ADHD-I/ADHD-HI	ALFF and ReHo	3 and 1	GRF $P<0.005$	MNI
Chen R, 2024 [35]	51	28	Adolescent	112.7±15.98	118.0±13.7	N, N/A	M	Siemens Trio 3.0 T	N/A	DSM-IV	ADHD-I/ADHD-C	fALFF	1	GRF $P<0.05$	MNI
Bin BL, 2024 [36]	33	25	Child	≥80	≥80	N, N/A	M and F	Siemens Vida 3.0 T	drug-naïve	DSM-5	N/A	ReHo	3	FDR $P<0.05$	MNI
Wei FQ, 2024 [37]	21	19	Child	≥80	≥80	N, N/A	M and F	Siemens Verio 3.0 T	stimulant-naïve	DSM-5	N/A	ALFF	2	GRF $P<0.05$	MNI
Hu R, 2025 [38]	48	44	Child	≥80	≥80	N/N	M and F	GE Healthcare 3.0 T	N/A	DSM-5	N/A	dALFF	6	FWE $P<0.05$	MNI

ADHD-I attention deficit hyperactivity disorder, inattentive type; *ADHD-HI, ADHD* hyperactive-impulsive type; *ADHD-C, ADHD* combined type; *ReHo* regional homogeneity; *ALFF* amplitude of low-frequency fluctuations; *fALFF* fractional ALFF; *dALFF* dynamic ALFF; *Puncor* uncorrected *p* value; *AlphaSim* AlphaSim correction; *GRF* Gaussian random field correction; *FDR* false discovery rate; *FWE* familywise error correction; *HCs* healthy controls; *M* male; *F* female; *N* not significant; *Y* significant; *N/A* not available; *MNI* Montreal Neurological Institute

replicated in 4 of 5 iterations. In contrast, clusters in the medial frontal gyrus were replicated in 3 of 5 iterations (Table S5).

A jackknife sensitivity analysis was not performed for increased brain activation in adolescents because of the limited number of studies ($n=4$), which was insufficient for a reliable estimation. A jackknife sensitivity analysis was also not conducted for increased brain activation in the child group or for any alterations in brain activity in the combined group of children and adolescents, as no significant results were identified.

Publication bias analysis

The included studies exhibited no significant risk of publication bias, as assessed by the NOS. Among these 28 studies, the overall risk of bias was low for 25 research items, with only 3 items having a moderate overall risk (Table S6). A total of 28 studies were assessed with the ROB 2 tool. Twenty were rated as low risk of bias in all domains, and eight had "some concerns" mainly in outcome measurement or result selection. Details are shown in Supplementary Fig. 1.

Discussion

This study is the first in which a meta-analysis based on previous rs-fMRI data was conducted to separately investigate functional alterations in the brains of children and adolescents with ADHD compared with healthy controls. The age-stratified analysis was preplanned to explore developmental differences between children and adolescents with ADHD. We analyzed changes in functional consistency within each age group. In children with ADHD, decreased activation was observed primarily in frontal regions associated with executive function and motor control, suggesting that frontal hypoactivation is a core neural feature in this age group. In adolescents with ADHD, altered activity was observed in both frontal and cerebellar regions, indicating disruptions in executive processes and motor coordination; this altered activity in adolescents reflects a more highly distributed pattern of network dysfunction than that in children. Combining children and adolescents with ADHD into a single group yielded no significant results. The contrast analysis revealed no overlapping clusters between children and adolescents with ADHD. While these results did not indicate statistically significant differences, the distinct patterns identified in age-stratified analyses suggest potential developmental differences in ADHD-related neural alterations. Collectively, these findings may reflect developmental dynamics in ADHD-related brain alterations across childhood and adolescence, suggesting that age-specific neurobiological markers could be valuable in guiding future research and informing targeted clinical interventions.

Increased spontaneous neural activity in children and adolescents with ADHD

Increased spontaneous brain activity was observed only in adolescents with ADHD, primarily within key regions of the sensorimotor network (SMN), including the paracentral lobule (BA 4 and 5), postcentral gyrus (BA 3), and medial frontal gyrus (BA 6). These areas are involved in voluntary motor control, somatosensory processing, and motor planning, respectively, and are crucial for integrating sensory input with motor output.

The enhanced spontaneous activity in these SMN regions may reflect atypical sensorimotor functioning commonly associated with ADHD symptoms such as hyperactivity and impulsivity. Earlier studies have reported disrupted SMN connectivity in ADHD patients during rs-fMRI, with these alterations correlating with symptom severity, including inattention, hyperactivity, and emotional dysregulation [55]. Moreover, recent findings suggest that the SMN in ADHD tends to exhibit prolonged low-entropy states and increased persistence, which may point to reduced network flexibility and impaired sensory regulation [56]. These alterations may reflect neurodevelopmental adaptations to the increasing demands of sensory processing and behavioral regulation during adolescence, suggesting a more complex interaction between neurodevelopment and behavioral outcomes in ADHD [57].

The paracentral lobule, situated at the intersection of the precentral and postcentral gyri, plays a crucial role in voluntary motor control and somatosensory processing [58]. In our analysis, adolescents with ADHD exhibited increased spontaneous activity in this region bilaterally. One potential explanation is that this may reflect heightened sensorimotor drive in response to increasing self-regulatory demands during adolescence [59]. While one structural study, Liang et al., have reported decreased structural covariance between the right cerebellar lobule VI and the left paracentral lobule in children with ADHD, implicating potential disruption of integrated sensorimotor circuits despite the absence of direct behavioral data [60]. Additionally, rs-fMRI evidence shows rightward lateralization in the paracentral lobule, suggesting atypical SMN organization possibly related to impaired motor control and attention regulation [61].

The postcentral gyrus, the primary somatosensory cortex [62], also showed increased spontaneous activity in the adolescents with ADHD. potentially reflecting abnormalities in sensory integration and regulation. This may reflect abnormal sensory responsivity, consistent with findings that

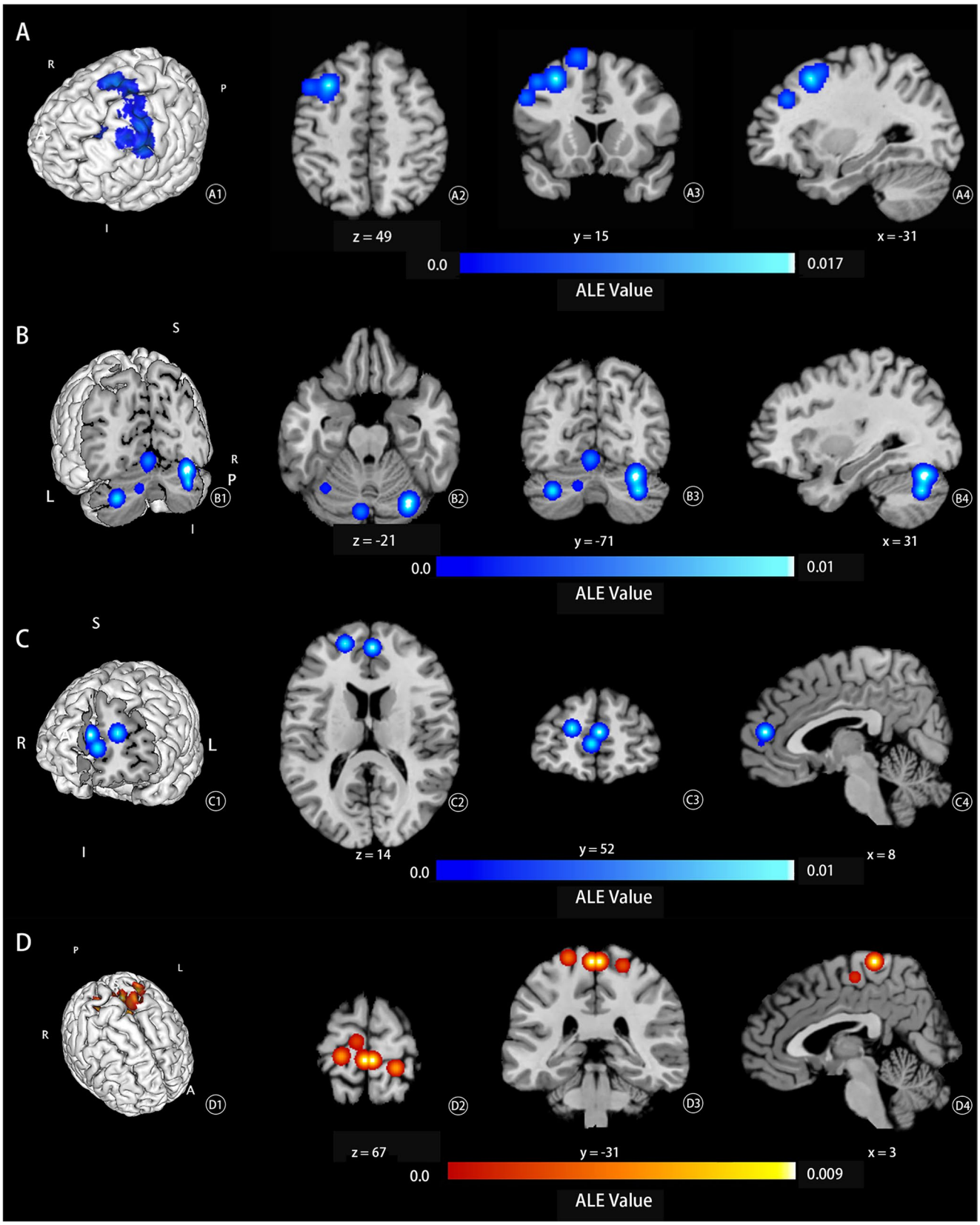

A
R
P
A1
I
A2
A3
A4
z = 49
y = 15
x = -31
0.0
0.017
ALE Value
B
S
R
P
L
B1
I
B2
B3
B4
z = -21
y = -71
x = 31
0.0
0.01
ALE Value
C
S
R
L
C1
I
C2
C3
C4
z = 14
y = 52
x = 8
0.0
0.01
ALE Value
D
P
L
R
A
D1
D2
D3
D4
z = 67
y = -31
x = 3
0.0
0.009
ALE Value

Fig. 2 Brain regions showing altered spontaneous activity in children and adolescents with ADHD compared with HCs. (**A**) Decreased spontaneous brain activity in children with ADHD compared with HCs (blue). (**B**, **C**) Decreased spontaneous brain activity in adolescents with ADHD compared with HCs (blue). (**D**) Increased spontaneous brain activity in adolescents with ADHD compared with HCs (red). The color bars indicate the ALE values. (A1, B1, C1, D1) Three-dimensional surface renderings of brain regions with altered spontaneous activity. (A2–A4, B2–B4, C2–C4, D2–D4) MNI coordinate slice views of the corresponding clusters. ADHD, attention deficit hyperactivity disorder; HCs, healthy controls; MNI, Montreal Neurological Institute; R, right; L, left

individuals with ADHD often experience hypersensitivity to tactile stimuli and dysregulated somatosensory feedback [63]. Functional connectivity studies further support this interpretation, indicating disrupted interactions between the postcentral gyrus and regions responsible for motor control and cognitive regulation [64, 65]. Moreover, Peter et al. found that abnormal connectivity involving the postcentral gyrus was linked to hyperactivity symptoms in ADHD, suggesting that sensorimotor dysfunction may persist beyond childhood into adulthood and contribute to core behavioral manifestations [66].

The medial frontal gyrus, a core region of the SMN, is involved in motor planning and sensorimotor integration [67]. In our study, adolescents with ADHD showed increased spontaneous activity in this region, potentially reflecting impaired motor regulation and impulsivity. Functional studies have linked medial frontal dysfunction to deficits in motor execution and attention [68], while structural findings indicate delayed cortical thinning in this area among adolescents with persistent ADHD symptoms [69]. Hu et al. further reported increased dynamic neural variability in the medial frontal gyrus in ADHD patients, which was associated with impairments in attention and executive functioning [38]. Moreover, Kumar et al. identified cortical thinning in the superior frontal gyrus and reduced gray matter in adjacent prefrontal areas in children with ADHD (ages 7.5 — 13), reinforcing the involvement of this broader frontal region in the disorder's pathophysiology [70]. Collectively, these findings highlight medial frontal abnormalities as central to the motor and executive dysfunctions observed in ADHD.

Taken together, the observed increases in SMN activity may reflect a combination of primary dysfunction and adaptive neurodevelopmental processes aimed at coping with the evolving demands of adolescence. However, due to the limited number of adolescent studies ($n=4$), jackknife sensitivity analysis could not be conducted for these results. Therefore, these findings should be interpreted with caution and warrant further validation in larger, age-specific samples.

Decreased spontaneous neural activity in children and adolescents with ADHD

In contrast to the increased spontaneous neural activity observed in adolescents, both children and adolescents with ADHD exhibited decreased spontaneous activity in distinct brain regions. These reductions were mainly localized to the frontal cortex in children and the cerebellar and medial prefrontal regions in adolescents, suggesting developmentally specific profiles of neural dysfunction.

In the child group, decreased spontaneous activity was located mainly in the left frontal lobe, including the middle frontal gyrus (BA 6 and 9), subgyral region (BA 6), superior frontal gyrus (BA 6), precentral gyrus (BA 9), and medial frontal gyrus (BA 6). These regions are central to the executive control network (ECN), which governs attention regulation, inhibitory control, and goal-directed behavior. Hypoactivity in this network aligns with well-established cognitive deficits in pediatric ADHD, including impulsivity and distractibility. However, jackknife sensitivity analysis showed moderate reproducibility (replicated in 9 out of 17 iterations), indicating variability across studies. This may reflect differences in study protocols, participant demographics, and imaging methods. In addition, strict motion exclusion criteria commonly employed in resting-state fMRI studies may have led to the systematic exclusion of younger children with more severe hyperactivity, who are more likely to move during scanning. As a result, the analytic samples may underrepresent this subgroup, potentially contributing to the limited jackknife sensitivity observed in child-specific results.

Despite this variability, our findings are consistent with prior evidence showing delayed cortical maturation, reduced gray matter, and weakened functional connectivity in the prefrontal cortex of children with ADHD [71]. Functionally, these regions form the ECN, responsible for attention regulation, inhibitory control, and goal-directed behavior [56]. Disruption of the ECN is a well-established neural basis for attentional lapses and executive dysfunction in pediatric ADHD. Recent studies confirm that children with ADHD have decreased connectivity within the frontoparietal and executive networks, and this weaker integration is associated with greater inattentive and impulsive symptoms [72, 73]. Evidence from multimodal neuroimaging studies provides additional support for this pattern. Liu et al. reported atypical functional connectivity between the middle frontal and precentral gyri, which may reflect immaturity in executive — motor integration [74]. Gonchigsuren et al. identified reduced cerebral blood flow and gray matter volume in overlapping frontal areas [75]. Lee et al. observed increased frontal theta activity using eLORETA, which may indicate cortical hypoarousal [76]. Hai et al. found

Table 2 Applying the ALE method to investigate alterations in functional activity in the brains of children with ADHD compared with HCs

Group comparison	Hemisphere	Anatomical label	Brodmann areas	X	Y	Z	ALE value	*P* value	Z score
Children with ADHD > HCs									
	-	-	-	-	-	-	-	-	-
Children with ADHD < HCs									
	L	middle frontal gyrus	BA 6	−30	16	48	0.01748562	6.5508243E-7	4.838223
	L	subgyrus, frontal lobe	BA 6	−20	10	60	0.009925162	2.3900135E-4	3.49279
	L	middle frontal gyrus	BA 9	−54	18	34	0.009377258	3.8485255E-4	3.3634653
	L	superior frontal gyrus	BA 6	−14	14	64	0.008980884	6.336125E-4	3.2233005
	L	middle frontal gyrus	BA 6	−44	16	46	0.008976641	6.336125E-4	3.2233005
	L	precentral gyrus	BA 9	−42	28	36	0.008886854	6.950115E-4	3.1967146
	L	middle frontal gyrus	BA 9	−32	34	32	0.008798306	7.663987E-4	3.1684012
	L	medial frontal gyrus	BA 6	−12	10	60	0.008268559	0.001209084	3.0333974
	L	superior frontal gyrus	BA 6	−10	22	68	0.007610812	0.002335764	2.8288522

ALE activation likelihood estimation; *ADHD* attention deficit hyperactivity disorder; *HCs* healthy controls; *BA* Brodmann area; *R* right; *L* left

Table 3 Applying the ALE method to investigate alterations in functional activity in the brains of adolescents with ADHD compared with HCs

Group comparison	Hemisphere	Anatomical label	Brodmann areas	Peak MNI coordinates			ALE value	*P* value	Z score
				X	Y	Z			
Adolescents with ADHD > HCs									
	L	paracentral lobule	BA 4	−4	−32	70	0.009326795	4.2735144E-5	3.9285216
	L	paracentral lobule	BA 4	2	−32	70	0.009317517	9.572286E-5	3.7300456
	L	medial frontal gyrus	BA 6	−4	−18	58	0.00805693	2.4988342E-4	3.4808815
	R	paracentral lobule	BA 5	20	−38	66	0.00805562	2.4988342E-4	3.4808815
	L	postcentral gyrus	BA 3	−20	−30	72	0.008033592	2.6542615E-4	3.4646854
	L	medial frontal gyrus	BA 6	−10	−18	74	0.007160132	7.047131E-4	3.1927137
Adolescents with ADHD < HCs									
	L	cerebellar tonsil	-	−32	−68	−36	0.009991506	1.2780817E-5	4.209787
	L	declive of the cerebellar posterior lobe	-	−6	−78	−20	0.008243072	1.8666594E-4	3.5582519
	L	anterior lobe of the cerebellum	-	−32	−60	−30	0.008153502	1.8932507E-4	3.554534
	L	declive of the cerebellar posterior lobe	-	−4	−72	−12	0.007417188	3.7059686E-4	3.37387
	L	nodule of the cerebellar anterior lobe	-	−14	−64	−32	0.007336426	3.8046922E-4	3.366626
	L	uvula of the cerebellar anterior lobe	-	30	−72	−26	0.011748782	3.7069226E-6	4.481456
	R	cerebellar tonsil	-	32	−70	−36	0.0101701	1.1212313E-5	4.2392764
	R	uvula of the cerebellar anterior lobe	-	18	−84	−32	0.008029219	2.739329E-4	3.4561918
	L	superior frontal gyrus	BA 10	−18	54	14	0.008325785	1.4618134E-4	3.6219752
	L	medial frontal gyrus	BA 10	−4	54	4	0.007775232	2.8369695E-4	3.4467404
	L	medial frontal gyrus	BA 9	2	52	12	0.007774996	3.5549392E-4	3.3853066

NA not applicable; *ALE* activation likelihood estimation; *ADHD* attention deficit hyperactivity disorder; *HCs* healthy controls; *BA* Brodmann area; *R* right; *L* left

decreased cortical thickness in the superior frontal gyrus, a finding potentially linked to symptoms of inattention and hyperactivity [77]. Although these studies differ in methodology and sample characteristics, they consistently highlight deviations in frontal development related to cognitive control in pediatric ADHD and emphasize the importance of age-specific interpretations of resting-state brain function in this population.

In adolescents with ADHD, decreased spontaneous activity was primarily observed in the cerebellum, including the tonsil, uvula, declive, anterior lobe, and nodule, as well as in the superior and medial frontal gyri (BA 9 and 10). These

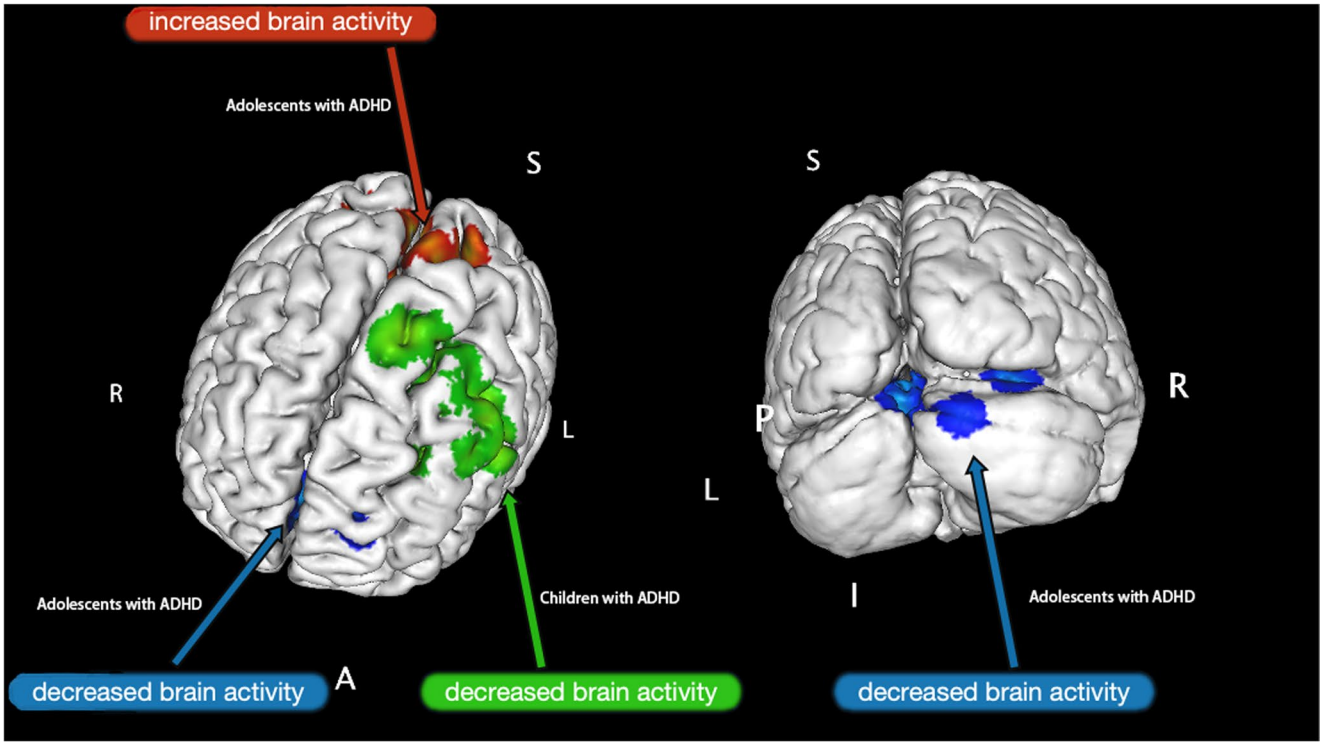


Fig. 3 Lack of spatial overlap in significantly altered brain clusters between children and adolescents with ADHD. The green regions represent brain regions with decreased spontaneous activity in children with ADHD compared with HCs. The blue regions represent brain regions with decreased spontaneous activity in adolescents with ADHD compared with HCs. The red regions represent brain regions with increased brain activity in adolescents with ADHD compared with HCs. All clusters are displayed on three-dimensional brain surface renderings. ADHD, attention deficit hyperactivity disorder; HCs, healthy controls; R, right; L, left; A, anterior; P, posterior; S, superior; I, inferior

Table 4 Applying the ALE method to investigate alterations in functional activity in the brains of children and adolescents with ADHD compared with HCs

Group comparison	Peak MNI coordinates	ALE value of dataset 1	ALE value of dataset 2	ALE value of the pooled dataset	ALE value of the contrast dataset
Children with ADHD>Adolescents with ADHD (increased activity)	-	0.01691	0.01085	0.0169	0.0
Children with ADHD<Adolescents with ADHD (decreased activity)	-	0.0175	0.0117	0.01748562	0.0

ALE activation likelihood estimation; *ADHD* attention deficit hyperactivity disorder; *HCs* healthy controls; dataset 1, increased brain functional activity in children with ADHD compared with that in HCs; dataset 2, increased brain functional activity in adolescents with ADHD compared with that in HCs

regions are involved in sensorimotor control, cognitive monitoring, and affective regulation, and are connected via cerebello-thalamo-cortical circuits that integrate motor and executive processes [79]. Dysregulation in these pathways may compromise the coordination of cognitive and motor outputs, contributing to hyperactivity and behavioral variability in adolescence [78].

However, the robustness of these findings was moderate. Jackknife analysis indicated relatively consistent cerebellar and superior frontal alterations (4/5 iterations), but less stable medial frontal findings (3/5), highlighting variability across studies and the need for cautious interpretation. Consistent with the present findings, several neuroimaging studies have identified structural and functional differences in the cerebellum among individuals with ADHD. Duan et al. reported an inverse relationship between gray matter volume in the cerebellar tonsil and inattention severity in adults, suggesting that cerebellar morphology may be associated with attentional processes [80]. Pagliaccio et al. observed reduced activation in the cerebellar tonsil and declive during cognitive load, which may reflect altered recruitment of posterior cerebellar regions during task engagement [81]. Yeh et al. found increased regional cerebral blood flow in the anterior cerebellar lobe in adolescents with ADHD and comorbid developmental coordination disorder, possibly indicating compensatory responses

to underlying sensorimotor or attentional difficulties [82]. Additionally, structural MRI results from Yu et al. revealed gray matter reductions in both the cerebellum and superior frontal gyrus in ADHD participants, pointing to broader cortico-cerebellar alterations [83].

The superior and medial frontal gyri (BA 9 and 10), core ECN nodes, are essential for sustained attention, error monitoring, and self-regulation, especially during adolescence. These regions also interact with the DMN, supporting dynamic switching between internally and externally focused cognition [84]. Prior neuroimaging evidence highlights their relevance to ADHD symptomatology. For example, Christakou et al. reported hypoactivation in the superior frontal gyrus during sustained attention tasks, correlating with inattention severity [85]. Similarly, Tan et al. found reduced anticorrelation between the medial frontal gyrus and the amygdala, suggesting disruptions in emotion–cognition integration and regulatory control [86].

Consistent with these prior findings, the present meta-analysis also revealed decreased spontaneous neural activity in the superior and medial frontal gyri in adolescents with ADHD. However, the spatial distribution and developmental context of this reduction in intrinsic activity differed markedly from those observed in children. In the child group, reduced activity was localized to more lateral and caudal frontal regions, including the middle frontal gyrus, precentral gyrus, and superior frontal gyrus (BAs 6 and 9), which are involved primarily in externally guided motor control and basic executive operations. This topographic shift suggests a developmental reorganization of frontal dysfunction, reflecting an age-related transition from deficits in externally guided motor and executive control to impairments in internally mediated functions, such as self-monitoring, goal maintenance, and behavioral regulation [87]. Despite some overlap in affected regions, the spatial divergence underscores that while executive dysfunction remains a core feature, its neural representation shifts with age, paralleling the increasing demands placed on self-directed cognition during adolescence [88]. These findings may point to a developmental shift in the neural architecture of ADHD—from focal frontal dysfunction in childhood to distributed network-level dysregulation in adolescence.

In summary, children with ADHD presented decreased spontaneous activity, particularly in lateral ECN regions, while adolescents exhibit decreased activity in medial prefrontal and cerebellar areas. These findings suggest an age-related shift in the distribution of neural dysfunction in ADHD. Recognizing these developmental differences is critical for interpreting resting-state alterations and tailoring age-specific interventions.

Developmental and clinical heterogeneity of alterations in the brains of children and adolescents with ADHD

The absence of overlapping clusters between children and adolescents in contrast analyses did not reveal statistically significant differences, but may still suggest possible age-specific neural mechanisms underlying ADHD. Reduced spontaneous frontal activity was predominant in children, whereas adolescents presented more highly distributed alterations involving both frontal and cerebellar regions. Although these results did not reach formal statistical significance in the direct contrast analysis, the distinct topographies observed in single-dataset analyses support the view that ADHD is not a static disorder but evolves with brain maturation and functional reorganization, as reflected in potential age-specific patterns of spontaneous neural activity corresponding to distinct stages of brain network development. Recent studies by Norman et al. highlight age-specific neural changes in ADHD. Increased rMFG connectivity linked to impulsivity, reduced DMN — task network segregation, and fronto-striatal dysconnectivity in adolescents reflect a developmental shift from focal frontal abnormalities in childhood to broader network dysfunction in adolescence, consistent with our findings [89–91].

Building on the observation of decreased spontaneous neural activity in frontal and cerebellar regions in children with ADHD, accumulating evidence highlights developmental differences in the neurobiological underpinnings of ADHD. During typical development, increasing segregation and specialization of large-scale functional networks—particularly the ECN, DMN, and SMN—enable more efficient cognitive flexibility, attentional control, and motor regulation [92, 93]. In ADHD, this process appears delayed or disrupted. Sripada et al. reported immature segregation between the DMN and VAN [94], while Gao et al. found increased frontoparietal network (FPN) — DMN connectivity and altered VAN interactions in children and adolescents with ADHD, reflecting atypical integration of executive and attention systems [95]. Liu et al. further showed that functional connectivity patterns evolve with age: children exhibit reduced fronto-subcortical connectivity, whereas adolescents show increased frontal cortical connectivity, suggesting a shift from underconnectivity to adaptive hyperconnectivity during maturation [74]. These patterns are consistent with our findings: children with ADHD showed localized reductions in spontaneous activity within prefrontal and motor regions, whereas adolescents exhibited more distributed changes, particularly in the FPN and DMN [94]. This shift may reflect adaptations to changing cognitive and behavioral demands, though its precise functional significance remains to be clarified.

Previous ALE meta-analyses of patients with ADHD have often pooled data across wide age ranges, potentially obscuring important developmental differences in brain function and masking age-specific neural signatures of the disorder [96]. In contrast, the present ALE meta-analysis independently examined children and adolescents, providing emerging evidence that ADHD is not a unitary neurobiological condition across development but instead reflects dynamic, developmentally specific alterations in brain activity. The absence of overlapping clusters in the contrast analysis, together with the distinct topography of altered spontaneous neural activity in children and adolescents, highlights the necessity of age-stratified approaches in ADHD neuroimaging research.

Notably, increased spontaneous neural activity was observed exclusively in adolescents with ADHD, but this finding could not be subjected to a jackknife sensitivity analysis because the number of contributing studies was insufficient. This asymmetry may reflect the relative paucity of neuroimaging studies specifically targeting adolescents. From a clinical perspective, the emergence of increased spontaneous activity in this age group may correspond to shifts in ADHD symptom expression during development. Hyperactivity and impulsivity tend to decrease with age, while inattentive symptoms often persist or intensify [97, 98]. Within the NIMH's Research Domain Criteria (RDoC) framework, these symptom dimensions are thought to reflect disruptions in distinct neural domains—sensorimotor and positive valence systems for hyperactivity/impulsivity and cognitive control systems for inattention. Rosch et al. further emphasized that such domain-specific disruptions are likely to follow sex- and age-dependent neurodevelopmental trajectories, particularly in intrinsic functional connectivity patterns involving fronto-subcortical circuits [99]. Thus, the adolescent-specific findings may reflect underlying changes in neurofunctional demands related to evolving clinical presentations. Moreover, adolescence is characterized by heightened interindividual variability in both symptom expression and neural development, potentially influenced by pubertal maturation, social-affective reorganization, and increasing executive load [100]. These factors underscore the importance of the developmental context when interpreting spontaneous neural activity in ADHD patients and suggest the need for more age-stratified, symptom-informed analyses.

Additionally, sex-related differences in brain development may further contribute to heterogeneity in neural alterations. Carucci et al. observed sex-specific neurodevelopmental trajectories in ADHD patients, with males showing more prominent disruptions in frontal regions and females exhibiting greater alterations in posterior cortical areas [101]. As approximately half of the studies in our meta-analysis included only male participants, the findings may disproportionately reflect male-biased patterns of neural dysfunction. This sex imbalance may obscure important differences in how ADHD manifests across sexes at the neural level. This finding highlights the need for future investigations to incorporate sex-stratified designs and examine sex as a potential moderator of neural heterogeneity in individuals with ADHD.

Finally, future studies should examine how the pubertal stage, rather than chronological age alone, may modulate spontaneous neural activity in ADHD patients. However, this interpretation remains tentative, and further longitudinal and multimodal studies are needed to clarify whether these differences reflect true neurodevelopmental progression or age-specific phenotypic variation.

Limitations

Despite the robustness of the present findings, several limitations merit consideration. First, the number of studies examining the adolescent group was relatively small, which limited the statistical power and may have affected the stability of the age-specific results. Moreover, variability in age and IQ across studies may have further introduced bias or reduced comparability. Second, most studies were conducted in Asian populations with predominantly Han ethnicities, and some studies included only male participants or did not distinguish ADHD subtypes. These factors may reduce the applicability of the current findings, given the potential neurobiological variations across sex, culture, and clinical presentation. Third, the included studies often adopted strict motion exclusion criteria to reduce noise, which may have resulted in the exclusion of individuals with more severe hyperactivity symptoms. As such, the neural mechanisms identified in this meta-analysis may primarily reflect those of participants who were able to maintain relatively low head motion during scanning. Consequently, the analyzed sample may be biased toward individuals with milder symptomatology or predominantly inattentive presentations. Therefore, our findings may overlook the extent or nature of spontaneous alterations in brain activity present in individuals with more pronounced hyperactivity and impulsivity. Fourth, although ALE effectively reduces false positives through spatial convergence, it may be less sensitive to false-negative results [102]. Finally, the present analysis was restricted to studies using ALFF and ReHo measures. While this criterion ensured methodological consistency, it excluded other informative resting-state approaches, such as seed-based connectivity or graph theory.

Conclusions

This study applied an age-stratified ALE meta-analytic approach to investigate spontaneous alterations in brain activity separately in children and adolescents with ADHD. The results revealed nonoverlapping profiles of neural dysfunction across development, characterized by focal reductions in spontaneous frontal activity in children with ADHD and more distributed frontal–cerebellar alterations in adolescents with ADHD. The identification of these age-specific neural markers may contribute to more tailored diagnostic frameworks and intervention strategies. The present neuroimaging findings may help refine neurodevelopmental models of ADHD and inform more individualized, developmentally sensitive strategies for diagnosis and intervention.

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Data availability The original contributions presented in this study are included in the article/Supplementary Material, and further inquiries can be directed to the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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References

1. Ahlberg R, Garcia-Argibay M, Rietz ED, Butwicka A, Cortese S, D'Onofrio BM, Ludvigsson JF, Larsson H (2023) Associations between Attention-Deficit/Hyperactivity disorder (ADHD), ADHD medication, and shorter height: a quasi-experimental and family-based study. J Am Acad Child Adolesc Psychiatry 62:1316–1325. https://doi.org/10.1016/j.jaac.2023.03.015
2. Baji I, Túri A, Nagy DL, Sterczer A (2024) Attention deficit hyperactivity disorder (ADHD) syndrome across ages. Dev Health Sci 6(2):34–38. https://doi.org/10.1556/2066.2023.00050
3. Faraone SV, Bellgrove MA, Brikell I et al (2024) Attention-deficit/hyperactivity disorder [published correction appears in Nat Rev Dis Primers. ;10(1):29. doi: 10.1038/s41572-024-00518-w.]. Nat Rev Dis Primers 10(1):11. Published 2024 Feb 22. https://doi.org/10.1038/s41572-024-00495-0
4. Idiazábal Alecha MÁ, Palau Baduell M (2025) Abordaje neurofisiológico En El Trastorno Por Déficit de Atención Con Hiperactividad [Neurophysiological approach to attention deficit hyperactivity Disorder]. Med (B Aires) 85(Suppl 1):42–47
5. Bedford SA, Lai MC, Lombardo MV et al (2025) Brain-charting autism and attention-deficit/hyperactivity disorder reveals distinct and overlapping neurobiology. Biol Psychiatry 97(5):517–530. https://doi.org/10.1016/j.biopsych.2024.07.024
6. Hoogman M, Bralten J, Hibar DP et al (2017) Subcortical brain volume differences in participants with attention deficit hyperactivity disorder in children and adults: a cross-sectional mega-analysis [published correction appears in Lancet Psychiatry 4(6):436. doi: 10.1016/S2215-0366(17)30213-4.]. Lancet Psychiatry 4(4):310–319. https://doi.org/10.1016/S2215-0366(17)30049-4
7. Ahmadi A, Saadatmand M, Wallois F (2024) Evaluation of potential alterations related to ADHD in the effective connectivity between the default mode network and cerebellum, hippocampus, thalamus, and primary visual cortex. Cereb Cortex 34(8):bhae335. https://doi.org/10.1093/cercor/bhae335
8. Mohammadkhanloo M, Pooyan M, Sharini H, Yousefpour M (2025) Investigating resting-state functional connectivity changes within procedural memory network across neuropsychiatric disorders using fMRI. BMC Med Imaging 25(1):18. https://doi.org/10.1186/s12880-024-01527-7
9. Shu YP, Zhang Q, Hou YZ et al (2024) Multimodal abnormalities of brain structures in adolescents and young adults with major depressive disorder: an activation likelihood estimation meta-analysis. World J Psychiatry 14(7):1106–1117. https://doi.org/10.5498/wjp.v14.i7.1106
10. Tian P, Zhu X, Liu Z et al (2025) Effects of vitamin D on brain function in preschool children with autism spectrum disorder: a resting-state functional MRI study. BMC Psychiatry 25:198. https://doi.org/10.1186/s12888-025-06534-8
11. Cao Q, Zang Y, Wang Y (2007) Brain functions in attention deficit hyperactivity disorder combined and inattentive subtypes: a resting-state functional magnetic resonance imaging study. Beijing Da Xue Xue Bao 39(3):261–265
12. Zang YF, He Y, Zhu CZ et al (2012) Altered baseline brain activity in children with ADHD revealed by resting-state functional MRI [published correction appears in Brain Dev 34(4):336]. Brain Dev 29(2):83–91. https://doi.org/10.1016/j.braindev.2006.07.002
13. Pang GF (2009) Cognitive development and its clinical application in normal children: a resting-state fMRI study. (Master's thesis)

14. Yang H, Wu QZ, Guo LT et al (2011) Abnormal spontaneous brain activity in medication-naïve ADHD children: a resting state fMRI study. Neurosci Lett 502(2):89–93. https://doi.org/10.1016/j.neulet.2011.07.028
15. Yang ZL (2011) Brain function in children with attention-deficit hyperactivity disorder: a resting-state fMRI study. Master's Thesis
16. Wu DQ (2012) Resting-state fMRI of ALFF and FC analysis in children with attention deficit hyperactivity disorder. Master's Thesis
17. An L, Cao QJ, Sui MQ et al (2013) Local synchronization and amplitude of the fluctuation of spontaneous brain activity in attention-deficit/hyperactivity disorder: a resting-state fMRI study. Neurosci Bull 29(5):603–613. https://doi.org/10.1007/s12264-013-1353-8
18. An L, Cao XH, Cao QJ et al (2013) Methylphenidate normalizes resting-state brain dysfunction in boys with attention deficit hyperactivity disorder. Neuropsychopharmacology 38(7):1287–1295. https://doi.org/10.1038/npp.2013.27
19. Yang RX, Ding KJ, Liu RX et al (2013) Study on attention-deficit hyperactivity disorder in children based on resting-state functional magnetic resonance imaging. Chin J Contemp Pediatr 15(9):723–727
20. Wang W (2013) A study of mechanism of ADHD based on ALFF and network topology. Master's Thesis
21. Li F, He N, Li Y et al (2014) Intrinsic brain abnormalities in attention deficit hyperactivity disorder: a resting-state functional MR imaging study. Radiology 272(2):514–523. https://doi.org/10.1148/radiol.14131622
22. Yang RX (2014) A resting-state brain function study of school-age children with attention-deficit/hyperactivity disorder based on functional magnetic resonance imaging. Master's Thesis
23. Wang SH (2014) Abnormalities of neural network in children with attention deficit hyperactivity disorder: a functional magnetic resonance imaging study. Doctor's Thesis
24. Yu X, Yuan B, Cao Q et al (2015) Frequency-specific abnormalities in regional homogeneity among children with Attention-Deficit/Hyperactivity disorder: A resting-state fMRI study. ADHD Atten Deficit Hyperactivity Disorders 7:36
25. Zhang XY, Yang C, Liang Y et al (2015) Different activities of amplitude of low-frequency fluctuation among attention deficit hyperactivity disorder subtypes in brain rs-fMRI. J Wenzhou Med Univ 45(02):100–105
26. Xie N, Sun LW, Xu SJ et al (2016) A study of brain structural and functional magnetic resonance imaging in children with attention deficit hyperactivity disorder. Chin J Magn Reson Imaging 7(08):582–586
27. Wang JB, Zheng LJ, Cao QJ et al (2017) Inconsistency in abnormal brain activity across cohorts of ADHD-200 in children with attention deficit hyperactivity disorder. Front Neurosci 11:320. https://doi.org/10.3389/fnins.2017.00320
28. Zhao D, Zheng S, Yang L et al (2017) Causal connectivity abnormalities of regional homogeneity in children with attention deficit hyperactivity disorder: a rest-state fMRI study. ADMET DMPK 5(4):242–252
29. Wang YF, Sun L, Liu L et al (2017) Sex differences in resting- state fractional amplitude of low frequency fluctuation of functional magnetic resonance imaging in children with attention- deficit / hyperactivity disorder. Chin Mental Health J 31(11):872–878
30. Kim JI, Yoo JH, Kim D, Jeong B, Kim BN (2018) The effects of GRIN2B and DRD4 gene variants on local functional connectivity in attention-deficit/hyperactivity disorder. Brain Imaging Behav 12(1):247–257. https://doi.org/10.1007/s11682-017-9690-2
31. Zhou JX, Gao YX, Bu X et al (2018) A multi-parameter resting-state functional magnetic resonance imaging study of brain intrinsic activity in attention deficit hyperactivity disorder children. J Biomed Eng 35(03):415–420
32. Lou F, Tao J, Zhou R et al (2021) Altered variability and concordance of dynamic resting-state fMRI indices in patients with attention deficit hyperactivity disorder. Front Neurosci 15:731596. https://doi.org/10.3389/fnins.2021.731596
33. Shang CY, Lin HY, Gau SS (2021) The norepinephrine transporter gene modulates intrinsic brain activity, visual memory, and visual attention in children with attention-deficit/hyperactivity disorder. Mol Psychiatry 26(8):4026–4035. https://doi.org/10.1038/s41380-019-0545-7
34. Feng Y, Zhu Y, Guo X et al (2023) Exploring the potential brain-cognition-behavior relationship in children with ADHD based on resting-state brain local activation and functional connectivity. J Atten Disord 27(14):1638–1649. https://doi.org/10.1177/10870547231197206
35. Chen R, Jiao Y, Zhu JS, Wang XH, Zhao MT (2024) Frequency-specific static and dynamic neural activity indices in children with different attention deficit hyperactivity disorder subtypes: a resting-state fMRI study. Front Hum Neurosci 18:1412572. https://doi.org/10.3389/fnhum.2024.1412572
36. Bin BL (2024) Resting-state functional magnetic resonance imaging study on brain spontaneous activity and functional networks in children with attention deficit hyperactivity disorder. Master's Thesis
37. Wei FQ (2024) Research on resting state functional magnetic resonance imaging of ADHD before and aftr executive function training based on virtual reality technology. Master's Thesis
38. Hu R, Du W, Tan F et al (2025) Dynamic alterations in spontaneous neural activity in patients with attention-deficit/hyperactivity disorder: a resting-state fMRI study. Brain Res Bull 222:111230. https://doi.org/10.1016/j.brainresbull.2025.111230
39. Costa T, Ferraro M, Manuello J et al (2024) Activation likelihood Estimation neuroimaging Meta-Analysis: a powerful tool for emotion research. Psychol Res Behav Manag 17:2331–2345
40. Samea F, Soluki S, Nejati V et al (2019) Brain alterations in children/adolescents with ADHD revisited: a neuroimaging meta-analysis of 96 structural and functional studies. Neurosci Biobehav Rev 100:1–8. https://doi.org/10.1016/j.neubiorev.2019.02.011
41. Lukito S, Norman L, Carlisi C et al (2020) Comparative meta-analyses of brain structural and functional abnormalities during cognitive control in attention-deficit/hyperactivity disorder and autism spectrum disorder. Psychol Med 50(6):894–919. https://doi.org/10.1017/S0033291720000574
42. Chen C, Sun S, Chen R et al (2025) A multimodal neuroimaging meta-analysis of functional and structural brain abnormalities in attention-deficit/hyperactivity disorder. Prog Neuropsychopharmacol Biol Psychiatry 136:111199. https://doi.org/10.1016/j.pnpbp.2024.111199
43. Sutcubasi B, Metin B, Kurban MK, Metin ZE, Beser B, Sonuga-Barke E (2020) Resting-state network dysconnectivity in ADHD: a system-neuroscience-based meta-analysis. World J Biol Psychiatry 21(9):662–672. https://doi.org/10.1080/15622975.2020.1775889
44. Cortese S, Aoki YY, Itahashi T, Castellanos FX, Eickhoff SB (2021) Systematic review and meta-analysis: resting-state functional magnetic resonance imaging studies of Attention-Deficit/Hyperactivity disorder. J Am Acad Child Adolesc Psychiatry 60(1):61–75. https://doi.org/10.1016/j.jaac.2020.08.014
45. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group (2009) Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med 6:e1000097. https://doi.org/10.1371/journal.pmed.1000097
46. Moreno MA, Binger K, Minich M, Zhao Q, Eickhoff J (2022) Adolescent digital technology interactions and importance:

associations with depression and Well-Being. Cyberpsychol Behav Soc Netw 25(4):245–249. https://doi.org/10.1089/cyber.2021.0294
47. Lo CK, Mertz D, Loeb M (2014) Newcastle-ottawa scale: comparing reviewers' to authors' assessments. BMC Med Res Methodol 14(1):45. https://doi.org/10.1186/1471-2288-14-45
48. Zhang Q, Hou YZ, Ding H et al (2024) Alterations of sleep deprivation on brain function: a coordinate-based resting-state functional magnetic resonance imaging meta-analysis. World J Psychiatry 14(2):315–329. https://doi.org/10.5498/wjp.v14.i2.315
49. Eickhoff SB, Laird AR, Grefkes C, Wang LE, Zilles K, Fox PT (2009) Coordinate-based activation likelihood estimation meta-analysis of neuroimaging data: a random-effects approach based on empirical estimates of spatial uncertainty. Hum Brain Mapp 30:2907–2926. https://doi.org/10.1002/hbm.20718
50. Zhang Q, Ding H (2024) Meta-analysis of resting-state fMRI in cervical spondylosis patients using AES-SDM. Front Neurol 15:1439939. https://doi.org/10.3389/fneur.2024.1439939
51. Eickhoff SB, Nichols TE, Laird AR et al (2016) Behavior, sensitivity, and power of activation likelihood estimation characterized by massive empirical simulation. Neuroimage 137:70–85. https://doi.org/10.1016/j.neuroimage.2016.04.072
52. Eickhoff SB, Bzdok D, Laird AR, Kurth F, Fox PT (2012) Activation likelihood estimation meta-analysis revisited. Neuroimage 59:2349–2361. https://doi.org/10.1016/j.neuroimage.2011.09.017
53. Lancaster JL, Tordesillas-Gutiérrez D, Martinez M et al (2007) Bias between MNI and Talairach coordinates analyzed using the ICBM-152 brain template. Hum Brain Mapp 28(11):1194–1205
54. Ding H, Zhang Q, Shu YP et al (2024) Vulnerable brain regions in adolescent major depressive disorder: a resting-state functional magnetic resonance imaging activation likelihood estimation meta-analysis. World J Psychiatry 14(3):456
55. Hua MH, Chen YL, Chen MH, Huang KL, Hsu JW, Bai YM, Wu YT (2021) Network-specific corticothalamic dysconnection in attention-deficit hyperactivity disorder. J Dev Behav Pediatr 42(2):122–127
56. Xin X, Gu S, Wang C, Gao X (2025) Abnormal brain entropy dynamics in ADHD. J Affect Disord 369:1099–1107. https://doi.org/10.1016/j.jad.2024.10.066
57. Fabio RA, Orsino C, Lecciso F, Levante A, Suriano R (2024) Atypical sensory processing in adolescents with attention deficit hyperactivity disorder: a comparative study. Res Dev Disabil 146:104674. https://doi.org/10.1016/j.ridd.2024.104674
58. Kimura Y, Yamada S, Komatsu K et al (2024) An anatomo-functional study of the interactivity between the paracentral lobule and the primary motor cortex. J Neurosurg 141(4):1096–1104. https://doi.org/10.3171/2024.2.JNS232753
59. Mennes M, Vega Potler N, Kelly C, Di Martino A, Castellanos FX, Milham MP (2012) Resting state functional connectivity correlates of inhibitory control in children with attention-deficit/hyperactivity disorder. Front Psychiatry 2:83. https://doi.org/10.3389/fpsyt.2011.00083
60. Liang S, Huang L, Zhan S et al (2023) Altered morphological characteristics and structural covariance connectivity associated with verbal working memory performance in ADHD children. Br J Radiol 96(1151):20230409. https://doi.org/10.1259/bjr.20230409
61. Zou H, Yang J (2021) Exploring the brain lateralization in ADHD based on variability of resting-state fMRI signal. J Atten Disord 25(2):258–264. https://doi.org/10.1177/1087054718816170
62. Alahmadi AAS (2024) Beyond boundaries: investigating shared and divergent connectivity in the pre-/postcentral gyri and supplementary motor area. Neuroreport 35(5):283–290. https://doi.org/10.1097/WNR.0000000000002011
63. Frost-Karlsson M, Capusan AJ, Olausson H, Boehme R (2024) Altered somatosensory processing in adult attention deficit hyperactivity disorder. BMC Psychiatry 24(1):558. https://doi.org/10.1186/s12888-024-06002-9
64. Agoalikum E, Klugah-Brown B, Wu H, Hu P, Jing J, Biswal B (2023) Structural differences among children, adolescents, and adults with attention-deficit/hyperactivity disorder and abnormal Granger causality of the right pallidum and whole-brain. Front Hum Neurosci 17:1076873. https://doi.org/10.3389/fnhum.2023.1076873
65. Fateh AA, Huang W, Mo T et al (2022) Abnormal insular dynamic functional connectivity and its relation to social dysfunctioning in children with attention deficit/hyperactivity disorder. Front Neurosci 16:890596. https://doi.org/10.3389/fnins.2022.890596
66. Sörös P, Hoxhaj E, Borel P et al (2019) Hyperactivity/restlessness is associated with increased functional connectivity in adults with ADHD: a dimensional analysis of resting state fMRI. BMC Psychiatry 19(1):43. https://doi.org/10.1186/s12888-019-2031-9
67. Jiang M, Yang S, Tan Y, Li X, He L (2024) Sensorimotor network ALFF markers as prognostic indicators of cervical spondylotic myelopathy post-decompression surgery outcomes. J Clin Neurosci 129:110769. https://doi.org/10.1016/j.jocn.2024.110769
68. Zhang H, Zhao Y, Cao W et al (2020) Aberrant functional connectivity in resting state networks of ADHD patients revealed by independent component analysis. BMC Neurosci 21(1):39. https://doi.org/10.1186/s12868-020-00589-x
69. Wang Y, Ma L, Wang J et al (2024) The neural and genetic underpinnings of different developmental trajectories of attention-deficit/hyperactivity symptoms in children and adolescents. BMC Med 22(1):223. https://doi.org/10.1186/s12916-024-03449-1
70. Kumar U, Arya A, Agarwal V (2017) Neural alterations in ADHD children as indicated by voxel-based cortical thickness and morphometry analysis. Brain Dev 39(5):403–410. https://doi.org/10.1016/j.braindev.2016.12.002
71. Wang Y, Zuo C, Xu Q, Hao L, Zhang Y (2021) Attention-deficit/hyperactivity disorder is characterized by a delay in subcortical maturation. Prog Neuropsychopharmacol Biol Psychiatry 104:110044. https://doi.org/10.1016/j.pnpbp.2020.110044
72. Hu Y, Ran J, Qiao R et al (2024) Identifying ADHD-related abnormal functional connectivity with a graph convolutional neural network. Neural Plast 2024:8862647. https://doi.org/10.1155/2024/8862647
73. Lin HY, Tseng WY, Lai MC, Matsuo K, Gau SS (2015) Altered resting-state frontoparietal control network in children with attention-deficit/hyperactivity disorder. J Int Neuropsychol Soc 21(4):271–284. https://doi.org/10.1017/S135561771500020X
74. Liu N, Liu Q, Yang Z et al (2023) Different functional alteration in attention-deficit/hyperactivity disorder across developmental age groups: a meta-analysis and an independent validation of resting-state functional connectivity studies. CNS Neurosci Ther 29(1):60–69. https://doi.org/10.1111/cns.14032
75. Gonchigsuren O, Harada M, Hisaoka S et al (2022) Brain abnormalities in children with attention-deficit/hyperactivity disorder assessed by multi-delay arterial spin labeling perfusion and voxel-based morphometry. Jpn J Radiol 40(6):568–577. https://doi.org/10.1007/s11604-021-01239-w
76. Lee I, Lee J, Lim MH, Kim KM (2021) Comparison of quantitative electroencephalography between tic disorder and Attention-Deficit/Hyperactivity disorder in children. Clin Psychopharmacol Neurosci 19(4):739–750. https://doi.org/10.9758/cpn.2021.19.4.739
77. Hai T, Swansburg R, Kahl CK et al (2022) Right superior frontal gyrus cortical thickness in pediatric ADHD. J Atten Disord 26(14):1895–1906. https://doi.org/10.1177/10870547221110918

78. Dieterich M, Brandt T (2024) Central vestibular networking for sensorimotor control, cognition, and emotion. Curr Opin Neurol 37(1):74–82. https://doi.org/10.1097/WCO.0000000000001233
79. Tao J, Jiang X, Wang X et al (2017) Disrupted control-related functional brain networks in drug-naive children with attention-deficit/hyperactivity disorder. Front Psychiatry 8:246. https://doi.org/10.3389/fpsyt.2017.00246
80. Duan K, Jiang W, Rootes-Murdy K et al (2021) Gray matter networks associated with attention and working memory deficit in ADHD across adolescence and adulthood. Transl Psychiatry 11(1):184. https://doi.org/10.1038/s41398-021-01301-1
81. Pagliaccio D, Wiggins JL, Adleman NE et al (2017) Behavioral and neural sustained attention deficits in disruptive mood dysregulation disorder and attention-deficit/hyperactivity disorder. J Am Acad Child Adolesc Psychiatry 56(5):426–435. https://doi.org/10.1016/j.jaac.2017.02.008
82. Yeh CB, Huang WS, Lo MC, Chang CJ, Ma KH, Shyu JF (2012) The rCBF brain mapping in adolescent ADHD comorbid developmental coordination disorder and its changes after MPH challenging. Eur J Paediatr Neurol 16(6):613–618. https://doi.org/10.1016/j.ejpn.2012.02.007
83. Yu M, Gao X, Niu X et al (2023) Meta-analysis of structural and functional alterations of brain in patients with attention-deficit/hyperactivity disorder. Front Psychiatry 13:1070142. https://doi.org/10.3389/fpsyt.2022.1070142
84. Kaboodvand N, Iravani B, Fransson P (2020) Dynamic synergetic configurations of resting-state networks in ADHD. Neuroimage 207:116347. https://doi.org/10.1016/j.neuroimage.2019.116347
85. Christakou A, Murphy CM, Chantiluke K et al (2013) Disorder-specific functional abnormalities during sustained attention in youth with attention deficit hyperactivity disorder (ADHD) and with autism [published correction appears in Mol Psychiatry 18(2):264]. Mol Psychiatry 18(2):236–244. https://doi.org/10.1038/mp.2011.185
86. Tan YW, Liu L, Wang YF et al (2020) Alterations of cerebral perfusion and functional brain connectivity in medication-naïve male adults with attention-deficit/hyperactivity disorder. CNS Neurosci Ther 26(2):197–206. https://doi.org/10.1111/cns.13185
87. Mastwal S, Li X, Stowell R et al (2023) Adolescent neurostimulation of dopamine circuit reverses genetic deficits in frontal cortex function. Elife 12:RP87414. https://doi.org/10.7554/eLife.87414
88. Sripada C, Kessler D, Fang Y, Welsh RC, Prem Kumar K, Angstadt M (2014) Disrupted network architecture of the resting brain in attention-deficit/hyperactivity disorder. Hum Brain Mapp 35(9):4693–4705. https://doi.org/10.1002/hbm.22504
89. Norman LJ, Sudre G, Price J, Shastri GG, Shaw P (2023) Evidence from big data for the default-mode hypothesis of ADHD: a mega-analysis of multiple large samples. Neuropsychopharmacology 48(2):281–289. https://doi.org/10.1038/s41386-022-01408-z
90. Norman LJ, Sudre G, Price J, Shaw P (2024) Subcortico-cortical dysconnectivity in ADHD: a voxel-wise mega-analysis across multiple cohorts. Am J Psychiatry 181(6):553–562. https://doi.org/10.1176/appi.ajp.20230026
91. Norman LJ, Sudre G, Bouyssi-Kobar M et al (2025) Cross-sectional mega-analysis of resting-state alterations associated with autism and attention-deficit/hyperactivity disorder in children and adolescents. Nat Ment Health 3:709–723. https://doi.org/10.1038/s44220-025-00431-5
92. Hoekzema E, Carmona S, Ramos-Quiroga JA et al (2014) An independent components and functional connectivity analysis of resting state fMRI data points to neural network dysregulation in adult ADHD. Hum Brain Mapp 35(4):1261–1272. https://doi.org/10.1002/hbm.22250
93. Metin B, Krebs RM, Wiersema JR et al (2015) Dysfunctional modulation of default mode network activity in attention-deficit/hyperactivity disorder. J Abnorm Psychol 124(1):208–214. https://doi.org/10.1037/abn0000013
94. Fair DA, Posner J, Nagel BJ et al (2010) Atypical default network connectivity in youth with attention-deficit/hyperactivity disorder. Biol Psychiatry 68(12):1084–1091. https://doi.org/10.1016/j.biopsych.2010.07.003
95. Gao Y, Shuai D, Bu X et al (2019) Impairments of large-scale functional networks in attention-deficit/hyperactivity disorder: a meta-analysis of resting-state functional connectivity. Psychol Med 49(15):2475–2485. https://doi.org/10.1017/S003329171900237X
96. Shu YP, Zhang Q, Li D, Liu JY, Wang XM, He Q, Hou YZ (2025) Vulnerable brain regions in adolescent attention deficit hyperactivity disorder: an activation likelihood estimation meta-analysis. World J Psychiatry 15(4):102215. https://doi.org/10.5498/wjp.v15.i4.102215
97. Pingault JB, Viding E, Galéra C et al (2015) Genetic and environmental influences on the developmental course of attention-deficit/hyperactivity disorder symptoms from childhood to adolescence. JAMA Psychiatr 72(7):651–658. https://doi.org/10.1001/jamapsychiatry.2015.0469
98. Hong SB, Hwang S (2022) Resting-state brain variability in youth with Attention-Deficit/Hyperactivity disorder. Front Psychiatry 13:918700. https://doi.org/10.3389/fpsyt.2022.918700
99. Rosch KS, Mostofsky SH, Nebel MB (2018) ADHD-related sex differences in fronto-subcortical intrinsic functional connectivity and associations with delay discounting. J Neurodev Disord 10(1):34. https://doi.org/10.1186/s11689-018-9254-9
100. Liang X, Sun L, Liao X et al (2024) Structural connectome architecture shapes the maturation of cortical morphology from childhood to adolescence. Nat Commun 15(1):784. https://doi.org/10.1038/s41467-024-44863-6
101. Carucci S, Narducci C, Bazzoni M et al (2023) Clinical characteristics, neuroimaging findings, and neuropsychological functioning in attention-deficit hyperactivity disorder: sex differences. J Neurosci Res 101(5):704–717. https://doi.org/10.1002/jnr.25038
102. Xiao Y, Dong S, Pan C et al (2024) Effectiveness of non-invasive brain stimulation on depressive symptoms targeting prefrontal cortex in functional magnetic resonance imaging studies: a combined systematic review and meta-analysis. Psychoradiology 4:kkae025